



**ARKA JAIN
University**
Jharkhand

**Department of Engineering
(B.Tech)**

ASSIGNMENT NO. - 03

NAME :- Devanshu Mahato

ENROLLMENT NO :- ASU/230355

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SUBMITTED TO :- Mr. Sushil Hasa

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FACULTY SIGNATURE

Apply Maclaurin's theorem to obtain the expression of $\log(1 + \sin x)$?

Sol $\Rightarrow f(x) \Rightarrow \log(1 + \sin x)$

$f(0) \Rightarrow \log(1+0)$

$f'(x) \Rightarrow \frac{\cos x}{1 + \sin x}$

$f'(0) \Rightarrow \frac{1}{1+0} = 1$

$f''(x) \Rightarrow \frac{1}{1 + \sin x}$

$f''(0) \Rightarrow \frac{1}{1+0} \Rightarrow 1$ (1)

$f'''(x) \Rightarrow \frac{\cos x}{(1 + \sin x)^2}$

$f'''(0) \Rightarrow \frac{1}{(1+0)^2} = 1$

$f^{(4)}(x) \Rightarrow \frac{-1 + \sin x + \cos^2 x}{(1 + \sin x)^3}$

$f^{(4)}(0) \Rightarrow \frac{-1 + 0 + 1}{(1+0)^3} \Rightarrow 0$ (2)

By Maclaurin's theorem

$\log(1 + \sin x) \Rightarrow f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \frac{f'''(0)}{6}x^3 + \dots$

$\Rightarrow 0 + 1 \cdot x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{2}{24}x^4 + \dots$

$\Rightarrow x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{12} + \dots$

$\therefore \log(1 + \sin x) \Rightarrow \left\{ x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{12} \right\}$

2) expand $\log(1+x)$

Sol $\Rightarrow$ let $f(x) \Rightarrow \log(1+x) \Rightarrow$ (1)

$f(0) \Rightarrow \log(1+0) \Rightarrow \log(1)$

$f'(x) \Rightarrow \frac{1}{1+x} \Rightarrow$ (2)

$$y'(0) \rightarrow \frac{1}{1+0}$$

$$f'(0) \rightarrow \frac{1}{1}$$

$$f'(0) \rightarrow \frac{1}{1}$$

$$f'(0) \rightarrow \frac{-1}{(1+x)^2} \dots \textcircled{10}$$

$$f'(0) \rightarrow -1$$

$$f'''(x) \rightarrow \frac{-(-2)(1+x)}{(1+x)^2} \dots \textcircled{12}$$

$$f'''(0) \rightarrow \frac{-(-2)}{1} \rightarrow \frac{2}{1} = 2$$

By Maclaurin's theorem

$$y(x) \rightarrow y(0) + x f'(0) + \frac{x^2}{2!} y''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$\log(1+x) \rightarrow x + \frac{x^2}{2!} + \frac{(2)x^3}{3!} + \dots$$

$$\log(1+x) \rightarrow x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4}$$

3) find the volume of the solid formed by the revolution of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Soln $\rightarrow$ Find the eqn of curve

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \dots \textcircled{1}$$

$$\frac{y^2}{b^2} \rightarrow 1 - \frac{x^2}{a^2}$$

$$\frac{y^2}{b^2} \rightarrow \frac{(a^2 - x^2)}{a^2}$$

$$y^2 \rightarrow \frac{b^2}{a^2} (a^2 - x^2) \dots \textcircled{2}$$

The required volume

$$2 \int_0^a x y^2 dx$$

$$\Rightarrow 2\pi \int_0^a \frac{b^2}{a^2} (a^2 - x^2) dx$$

$$\Rightarrow 2\pi \frac{b^2}{a^2} \int_0^a (a^2 - x^2) dx$$

$$\Rightarrow \frac{2\pi b^2}{a^2} \left[\int_0^a a^2 dx - \int_0^a x^2 dx \right]$$

$$\Rightarrow \frac{2\pi b^2}{a^2} \left[a^2 \int_0^a 1 dx - \left(\frac{x^3}{3} \right)_0^a \right]$$

$$\Rightarrow \frac{2\pi b^2}{a^2} \left[a^2 (x)_0^a - \left(\frac{a^3}{3} - \frac{0^3}{3} \right) \right]$$

$$\Rightarrow \frac{2\pi b^2}{a^2} \left[a^2 (a - 0) - \frac{a^3}{3} \right]$$

$$\Rightarrow \frac{2\pi b^2}{a^2} \left[a^3 - \frac{a^3}{3} \right]$$

$$\Rightarrow \frac{2\pi b^2}{a^2} \left[\frac{3a^3 - a^3}{3} \right]$$

$$\Rightarrow \frac{2\pi b^2}{a^2} \times \frac{2a^3}{3}$$

$$\Rightarrow \frac{4}{3} \pi a b^2$$

4) find the maximum and minimum values of the function.
 $f(x, y) = x^3 + y^3 - 3x - 12y + 12$

Solⁿ:

$$f(x, y) = x^3 + y^3 - 3x - 12y + 12$$

partial derivative

$$\frac{dF}{dx} \Rightarrow 0 + 3x - 0 - 3 \times 1 + 0$$

$$\frac{dF}{dx} \Rightarrow 3x^2 - 3$$

$$\frac{\partial F}{\partial y} \Rightarrow (3y^2 - 12)$$

for maxima and minima

$$\frac{dF}{dx} = 0, \quad \frac{dF}{dy} = 0$$

$$3x^2 - 3 = 0, \quad 3y^2 - 12 = 0$$

$$\{x = \pm 1\}, \quad \{y = \pm 2\}$$

Critical points $\rightarrow (1, 2), (1, -2), (-1, 2), (-1, -2)$

Evaluate the function at critical points

$$f(1, 2) \rightarrow 1 + 8 - 3 - 24 + 10 \rightarrow -8$$

$$f(1, -2) \rightarrow 1 - 8 - 3 + 24 + 10 \rightarrow 24$$

$$f(-1, 2) \rightarrow -1 + 8 + 3 - 24 + 10 \rightarrow (-4)$$

$$f(-1, -2) \rightarrow -1 - 8 + 3 + 24 + 10 \rightarrow (28)$$

So, the minimum value $\rightarrow (-8)$
 minimum value $\rightarrow (28)$

5) Verify Lagrange mean value theorem for $y(x) = (x-1)(x-2)(x-3)$ in $(0, 4)$.

$$\text{Sol} \rightarrow f(x) \rightarrow (x-1)(x-2)(x-3)$$

$$\rightarrow (x-1)(x^2 - 5x + 6)$$

$$\rightarrow x^3 - 5x^2 + 6x - x^2 + 5x - 6$$

$$\rightarrow x^3 - 6x^2 + 11x - 6$$

which is a polynomial function

hence, it is continuous on $[0, 4]$ and differentiable

on $(0, 4)$ thus, the function satisfies the conditions of Lagrange's mean value theorem.

hence exists $c \in (0, 4)$ such that

$$f'(c) \rightarrow \frac{f(4) - f(0)}{4 - 0} \quad \dots \textcircled{1}$$

Now,

$$f(x) \rightarrow (x-1)(x-2)(x-3)$$

$$f(0) \rightarrow (0-1)(0-2)(0-3)$$

$$-1) (-2) (-3)$$

$$1 - 6$$

$$f(4) = (4-1)(4-2)(4-3)$$

$$= 3 \cdot 2 \cdot 1$$

$$= 6$$

$$f'(x) = \frac{d}{dx} (x^3 - 6x^2 + 11x - 6)$$

$$= 3x^2 - 12x + 11$$

$$= 3x^2 - 12x + 11$$

$$\{ f'(c) = 3c^2 - 12c + 11 \}$$

from (1)

$$3c^2 - 12c + 11 = 6 - \frac{(c-6)}{4}$$

$$3c^2 - 12c + 11 = 3$$

$$3c^2 - 12c + 8 = 0$$

$$c = \frac{12 \pm \sqrt{144 - 4(3)(8)}}{2(3)}$$

$$c = \frac{12 \pm \sqrt{48}}{6} = \frac{12 \pm 4\sqrt{3}}{6} = 2 \pm \frac{2}{\sqrt{3}} \in (0, 4]$$

6) verify the Rolle's theorem for $f(x) = 3x^4 - 4x^2 + 5$ in interval $[1, 7]$

i) $f(x) = 3x^4 - 4x^2 + 5$ is a polynomial

$f(x)$ is continuous on $[1, 7]$

ii) since $f(x)$ is poly

$f'(x) = 12x^3 - 8x$, $f(x)$ is differentiable in $[1, 7]$

$$iii) f(1) = 3(1)^4 - 4(1)^2 + 5$$

$$f(1) = 4$$

$$f(7) = 3(7)^4 - 4(7)^2 + 5$$

$$f(1) = f(7)$$

Hence Rolle's theorem is verified

7) Trace the curve $x^3 + y^3 = 3axy$

Let's find

$$x^3 + y^3 = 3axy \quad \dots (1)$$

Symmetry

By interchanging x and y in eq (1) we get,

$$y^3 + x^3 = 3ayx$$

$$x^3 + y^3 = 3axy$$

The curve is symmetrical about the line $(y=x)$.

ii) Origin :

There is no constant term in the eq of the curve so the curve passes through the origin.

iii) Tangents at origin :

Tangents at origin are given by

$$3axy = 0$$

$$xy = 0$$

$$x=0, y=0$$

iv) Point of intersection

put $y=x$ in eq (1)

$$x^3 + x^3 = 3ax^2$$

$$2x^3 - 3ax^2 = 0$$

$$x^2 (2x - 3a) = 0$$

$$x^2 = 0, \quad 2x - 3a = 0$$

$$x = 0, \quad x = \frac{3a}{2}$$

$$(0,0), \quad \left(\frac{3a}{2}, \frac{3a}{2}\right)$$

Asymptotes:

Let $Q(x, y) = x^3 + y^3$

$Q_n(x, y) = -3axy$

Let $y = mx + c$

$(x=1), (y=m)$

$Q_n(m) = 1 + m^3, Q_n(m) = 0$

$1 + m^3 = 0$

$(m = -1)$

$c = \frac{Q_n + 1(m)}{Q_n(m)} = \frac{-3am}{3m^2} = \frac{-a}{m} = \frac{-a}{-1} = (-a)$

$y = -x - a$

$(x + y = -a)$

$(x=0) \mid y=0$
 $y=-a \mid x=-a$

$\therefore (-a, 0)$

vi) tangent at $(\frac{3a}{2}, \frac{3a}{2})$

$x = x + \frac{3a}{2}, y = y + \frac{3a}{2}$

$(x + \frac{3a}{2})^3 + (y + \frac{3a}{2})^3 - 3a(x + \frac{3a}{2})(y + \frac{3a}{2}) = 0$

$\{ x+y = 0 \}$

∴ Since the curve $x^2/3 + y^2/3 = a^2/3$ also find the whole area.

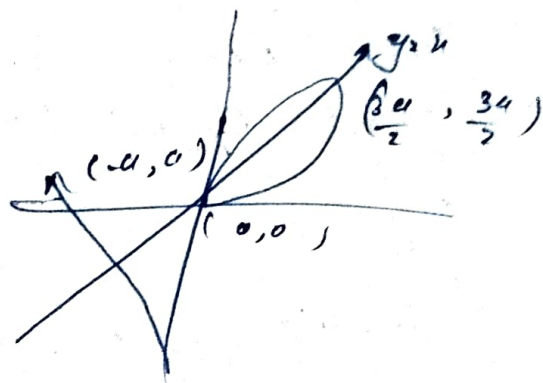
Sol 2.)

Given eqn of curve is .)

$\{ x^2/3 + y^2/3 = a^2/3 \}$

Symmetry.)

If we put $(x = -x)$ the eq of curve does not change. and if we put $y = -y$ the eq of curve does change.



So, the curve is symmetrical about x -axis and y -axis

Also, if we interchange x and y the eq of curve remains unchanged, so curve is symmetrical about the line $(y=x)$

Origin

Putting $x=0, y=0$ in LHS of given curve
 $0^{2/3} + 0^{2/3}$
 $0 \neq a^{2/3}$

So we have not passed through origin
Point of intersection

Putting $x=0$

we get, $y^{2/3} = a^{2/3}$

$$y = \pm a$$

the curve cuts the y -axis at $(0,0)$ and $(0, \pm a)$ also, putting $(y=0)$

$$x^{2/3} = a^{2/3}$$

$$(x = \pm a)$$

So, the curve cuts the x -axis at $(0,0)$

Region: if $x > 0$ and $x < -a$ then

$$y^{2/3} = (-x)^{2/3}$$

y will be imaginary

There is no portion of the curve in the
RHS of $x=a$ and LHS of $x=0$

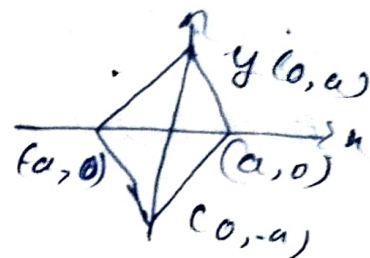
Given curve is

$$x^{2/3} + y^{2/3} = a$$

$$2) \int_0^a y \, dx$$

$$\Rightarrow \int_0^a (a^2/3 - x^2/3)^{3/2} dx$$

$$\Rightarrow \int_0^{\pi/2} (a \sin 3\theta) (-3a \cos^2 \theta)$$



Let $x = a \cos^2 \theta$

$$\Rightarrow \int 2a^2 \int_0^{\pi/2} \sin 4\theta \cos^2 \theta \, d\theta$$

$$\Rightarrow 12a^2 \times \frac{3}{6} \times \frac{1}{4} \times \frac{1}{2} \times \frac{\pi}{2}$$

$$\Rightarrow \frac{3\pi a^2}{8} \text{ Sq. unit.}$$

3) Trace the curve $y^2(a-x) = x^2(a+x)$
Symmetric

The given curve is symmetric about both the axes since it has even powers of x and y .

Origin

$$y^2(a-x) = x^2(a+x)$$

put $(x=0) \text{ and } (y=0)$

the given curve is passing through $(0,0)$

Tangent at origin

equating the lowest degree term to zero

$$y^2(a-x) = x^2(a+x)$$

$$y^2 a - x^2 a = x^2 y^2 + x^3$$

$$(y^2 - x^2) a = x^2 y^2 + x^3$$

$$y^2 - x^2 = 0$$

$$y^2 = x^2$$

$$y = \pm x$$

$y > 0$ are the tangent at origin

Asymptotes

parallel to (x -axis)

$$y^2(a-x) = x^2(a+x)$$

$$y^2 a - x y^2 = x^2 a + x^3 = 0$$

Since the co-efficients of highest powers of x and y are merely constant. Asymptote parallel to

$$y = (a-x) = x^2(a+x)$$

$$y^2 a - x y^2 = x^2 a + x^3$$

$$y^2(a-x) = x^2 a + x^3$$

Equate the co-efficient of the equal powers of y to 0
 $a-x=0$
 $x=a$ is the asymptote parallel to (y -axis)

points

$$y^2(a-x) = x^3(a+x)$$

$$\text{but } y=0$$

$$0 = x^2(a+x)$$

$$x=0$$

$$x=0$$

$$(0,0)$$

$$x+a=0$$

$$(x=-a)$$

$$(-a,0)$$

$$(-a,0)$$

$$\text{but } x=0$$

$$(x=0, y=0)$$

